

TETIO TSAGUE

JUNIOR SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Final-year Applied Computer Science student (BSc, expected July 2026) and junior software engineer with hands-on, full-stack delivery experience in **Java / Spring Boot, Python** and **TypeScript**. I design and build secure **REST APIs** (JWT, CSRF), relational data models and maintainable, pattern-driven object-oriented code. Delivered a heuristic decision-support tool during an internship at Infrabel and was the lead contributor on a multi-developer web platform. Solid foundations in algorithms, data structures, databases and computer networking. Rigorous, curious and security-minded — eager to contribute to SWIFT's mission of safe, reliable, worldwide financial messaging.

TECHNICAL SKILLS

Languages	Java, Python, TypeScript, JavaScript, C++, PHP, SQL
Backend & APIs	Spring Boot, FastAPI, REST API design, JWT authentication, CSRF protection, Maven, JPA / Hibernate
Frontend	React, Angular, Vite, Next.js, JavaFX, HTML5, CSS3, Tailwind CSS, responsive UI
Data & ML	PyTorch, MONAI, scikit-learn (k-NN, Naive Bayes), computer vision (OpenCV), data mining, dimensional modeling (star schema), data warehousing
Databases	MySQL, PostgreSQL, SQLite, relational design & SQL
Tools & Practices	Git, GitHub, Docker, Vercel, IntelliJ IDEA, Eclipse, unit testing, Agile / Scrum teamwork
Engineering	OOP & SOLID, design patterns (MVC, Factory, Iterator, command / CQRS-style), algorithms & data structures, networking (TCP/IP, subnetting, VLSM)
Spoken languages	French — native · English — professional working proficiency (B2)

PROFESSIONAL EXPERIENCE

Software Engineering Intern — Infrabel Feb 2026 – May 2026 · Brussels, Belgium
National railway infrastructure manager of Belgium

- Designed and built a **decision-support tool** that correlates operational alarms from the ICC department with planned maintenance interventions, using **heuristic matching rules** to surface the most likely alarm-to-intervention links.
- Cross-referenced large operational datasets to replace slow manual analysis, giving engineers a faster, evidence-based view for planning.
- Worked inside an established engineering team — from requirement analysis to a working, demonstrated tool — through iterative feedback cycles.

Academic Tutor — Software & Data Science (self-employed) Aug 2024 – Present · Brussels, Belgium

- Coach higher-education students in **data analytics and machine learning**: model design and evaluation (k-NN, Naive Bayes), dimensional modeling (fact & dimension tables), data warehousing, data mining and data analysis.
- Translate complex technical concepts into clear, structured explanations — measurably improving students' understanding and exam results.

SELECTED PROJECTS

Radixa — AI-Assisted Dental Diagnosis Platform Solo · R&D / MVP
Python (FastAPI) · Angular · PyTorch / MONAI · YOLOv8 / nnU-Net · Docker

End-to-end platform that detects dental pathologies on radiographs for both **adult and pediatric** dentition — an angle under-served by existing tools. Combines fine-tuned medical-imaging models (MONAI, YOLOv8, nnU-Net) with explainable predictions and calibrated confidence scores, served through a FastAPI inference API and an Angular practitioner UI. Privacy-by-design (GDPR / EU MDR) with local edge inference; the clinician always validates the AI's suggestions.

AEMT Notes — Full-Stack Knowledge Application Team · Exam project, 2026
React + TypeScript · Spring Boot (Java) · REST · JWT · Maven

Hierarchical note & folder application with secure **JWT authentication**, note linking and one-click **ZIP export** of an entire folder tree. Architected the Spring Boot backend around a clean, command-based (CQRS-style) *handler-per-use-case* structure; designed and shipped under timed exam conditions.

Movie Catalog & Recommendation Platform (ProjetTM) PHP · MySQL · JavaScript	Team · Lead contributor
Led development as the top contributor: implemented secure session authentication , CSRF-token protection , an admin moderation panel, bulk movie import, and a personalized recommendation feature backed by a MySQL database.	
Game World English — Multiplayer Vocabulary Game React · Vite · TypeScript · framer-motion · Serverless API (Vercel)	Solo
Offline-first English⇌French vocabulary game: on-device progress (localStorage), a light spaced-repetition engine that re-queues missed words, animated UI, and real-time multiplayer rooms (lobby, live scoreboard, QR-code join) powered by serverless functions.	
TTMC — Interactive Quiz Board Game Java · JavaFX	Team · OOP showcase
Multiplayer quiz board game on a clean MVC architecture with Factory (question creation) and Iterator (pawn traversal) design patterns and robust custom exception handling — a focused object-oriented engineering showcase.	
IP Subnetting & Network Planning Toolkit Python · SQLite	Team
Performs classful and VLSM subnet calculation and verification with automatic network-class detection, user authentication and SQLite persistence — used to plan and validate IP addressing schemes.	
Rubik's Cube Solver — Computer Vision Python · OpenCV · Solving algorithm	Solo
Recognizes the cube's state from images via computer vision (facelet & colour detection), then computes a valid solution and outputs the step-by-step move sequence.	
Other projects	Solo & Team
MamanMoi — community platform for young mothers (Spring Boot, JWT, feed / agenda / moderation) · Oneirica — 2D platformer game (C++).	

EDUCATION

BSc in Applied Computer Science — Application Development, Haute École Louvain en Hainaut (HELHa)	Sep 2023 – Jul 2026 (expected) · Mons, BE
Key coursework: software engineering & architecture, algorithms & data structures, databases & SQL, web & mobile development, computer networks, operating systems, object-oriented programming, machine learning & data analytics, IT security, project management (Agile).	
BSc Computer Science (first year), University of Yaoundé I	Sep 2022 – Jul 2023 · Cameroon
Advanced mathematics, electronics & physics, and programming foundations.	
Scientific Baccalaureate — Mathematics & Physics, Collège de La Retraite	2022 · Yaoundé, Cameroon
Intensive mathematics (5h/week), physics (4h/week) and chemistry.	

STRENGTHS & INTERESTS

Strengths: autonomy · technical curiosity · clear communication & mentoring · teamwork · problem-solving · attention to detail & a security-first mindset
Interests: logic puzzles & speedcubing (Rubik's Cube) · reading · sport · music